

Joel Bhattarai

joelbh@umich.edu | [248-974-4375](tel:248-974-4375) | [linkedin.com/in/joel-bhattarai](https://www.linkedin.com/in/joel-bhattarai) | East Palo Alto, California

EDUCATION

University of Michigan

Bachelor of Science in Engineering in Mechanical Engineering

Ann Arbor, MI

May 2026

EXPERIENCE

Automation Engineer

3D Infotech

Irvine, CA

August 2026 – Present

- Completed training in PolyWorks metrology and FANUC and Universal Robots scripting and operation.
- Implemented fully in-line and semi-in-line systems from design to integration, testing, and deployment.
- Gathered and analyzed metrology data for a variety of different geometries and applications.

Principal Investigator

NASA L'SPACE Program - Proposal Writing and Evaluation Experience

May 2025 – September 2025

- Independently generated a new technology within the areas of Autonomy, Robotics, and AI that was selected for a full proposal by a NASA panel.
- Led a group of ten interdisciplinary students to a full proposal and an NTR (New Technology Report).
- Advanced proposed technology from concept to final proposal.
- Served as the primary reviewer on a NASA proposal review board.

Lead Systems Engineer

NASA L'SPACE Program - Mission Concept Academy

May 2024 – September 2024

- Managed a top ten performing team of engineering students to a successful PDR.
- Designed a rover to accomplish practical missions on the surface of the moon.
- Independently developed system interface and wiring diagrams while managing overall rover metrics such as weight, power, and size.
- Completed skills modules on Heat Transfer, Siemens NX, Teaming, Engineering Requirements, Project Management, Systems Topics, Risk Management.

PROJECTS

Inverted Pendulum Stabilization

Automotive Laboratory

November 2025

University of Michigan

- Modeled physical behavior of inverted pendulum setup in state space for control design.
- Numerically determined multiple gain values using pole placement for a full-state feedback controller.
- Leveraged Simulink to apply solution to physical hardware on Quanser Qube Servo 3 and evaluated results based on real-world performance.
- Reduced settling time to less than 2 seconds with a maximum control input of 10 V.

SKILLS

Software & Simulation: PolyWorks, SolidWorks, Siemens NX, HyperMesh, HyperWorks, OptiStruct, ANSYS, LaTeX, Microsoft Azure AI Fundamentals Certified

Controls & Computing: MATLAB, Simulink, LabVIEW, Arduino, C/C++

Manufacturing & Lab: Machining (Mill & Lathe), Metalworking, Waterjetting, Geometric Dimensioning & Tolerancing (GD&T), Oscilloscopes, Multimeters, Strain Gauges, Thermocouples

COURSES

Relevant Coursework

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|----------------------------------|------------------------------|---------------------------|
| • Solid Mechanics and Statics | • Fluid Mechanics | • Automatic Control |
| • Thermodynamics | • Heat Transfer | • Energy Infrastructure |
| • Dynamics and Vibrations | • Control of Dynamic Systems | • Modern Energy Materials |
| • Design and Manufacturing I-III | • Behavior of Materials | |
| • Finite Elements | • Electric Circuits | |